Thinking Sideways, Observed: Interpreting the Lateral-Thinking Search Behavior of LLM Agents

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Complete draft; no open gaps.

Abstract

Interactive agents are usually evaluated by endpoint scores, yet a score records where a search ended, not how it moved. We study this gap in Turtle Soup, a lateral-thinking game in which an agent must reconstruct a hidden story through yes/no questions. We ran three Qwen models (4B, 27B, 397B) on 22 human-played puzzles, scoring a forced checkpoint answer every round and embedding each round's question to form an association trajectory. We found that: (1) the flat accuracy curve is not a resource limit: gains stop after round ten at every scale (397B $0.124 \rightarrow 0.200$, then flat; 4B flat at 0.05–0.06), longer budgets monotonically *hurt* the smallest model ($0.050 \rightarrow 0.004$), and the plateau is an equilibrium of finding and losing: models keep only 0.69–0.76 of the understanding they had held across two consecutive checkpoints, a retention scale leaves unchanged even as it raises the peaks 3.5-fold; (2) two behavioral patterns underlie this, read jointly from geometry and language: the small model *circles* (late-game stride 0.030 vs the mid-size model's 0.082; 2 of 66 games committed), whereas larger models *abandon*, reaching the solution mid-game and then failing to keep or commit it; (3) within puzzles, mixed-effects models find no predictive power for stride or drift (likelihood-ratio tests, $p \geq 0.20$), a conclusion unchanged under a surface-only anchor, so the geometry is an interpretive instrument rather than a score predictor; and (4) an unaudited Oracle collapses questions to "irrelevant" and makes every game unwinnable, so agent claims must begin with an information audit of the environment. We release the harness, a provenance-enforced puzzle set with an under-determination difficulty measure, and a two-part score whose clue-recall and causal-logic halves demonstrably measure different competences.

1 Introduction

Here is a Turtle Soup puzzle. *A man's body lies in the desert, clutching half a matchstick; luggage and clothing are scattered around him.* That is all the solver sees. The hidden story: a hot-air balloon was losing altitude; the passengers threw out their luggage, still could not stay up, and drew lots with matchsticks; the man who drew the short one was thrown out. The solver recovers this story by asking yes/no questions, one per round.

Outcome metrics can say whether an agent got there. On our verified set, that story is flat: accuracy rises only in the first ten of thirty rounds, only for the larger models, and twenty further rounds buy nothing at any scale we tested (§5). The failure is not a budget limit, and scaling does not remove it: scale raises the peaks a model reaches mid-game, but not the fraction it keeps. What outcome metrics cannot say is how the agent searched. Two agents scoring 0.0 on the same puzzle may be failing in opposite ways: one never leaves its first hypothesis, while the other finds the answer and lets it go. Telling them apart requires instruments that read the process; providing them is this paper's contribution.

Why Turtle Soup. The behavior *is* text, so no probing or activation access is needed; puzzles are deliberately under-determined, so each associative jump's direction is diagnostic; and games are short, language-only, and cheap at scale.

Central hypothesis. An agent's per-round questions, embedded in a semantic space, form a trajectory whose geometry explains success and failure. **H1 (stalling)**: an agent stuck in a hypothesis basin shows its stride (per-round semantic step size) shrinking toward zero, of which lexical novelty heuristics catch only the verbatim extreme. **H2 (drifting)**: an agent exploring where no human path goes shows normal stride but growing anchor distance, which lexical novelty misses entirely. **H3 (predictivity)**: trajectory features predict final solve quality beyond outcome-adjacent covariates. The results confirm H1 (as circling), reject H2's drift, reveal a second, unhypothesized pattern (abandoning), and do not support H3 (§5).

Contributions. (1) An explanation of the flat curve: task failure is driven neither by round budget nor by model capability alone, but by ineffective exploration; scale raises what a model momentarily finds, not what it keeps. (2) Two behavioral patterns, *circling* and *abandoning*, identified jointly in geometry space and in language space, with an honest account of what the geometry does and does not predict. (3) An open-source harness, a **22-puzzle set restricted to puzzles humans have played and solved**, and a composite score decoupling clue recall from judged causal logic. (4) A puzzle-level difficulty measure, **under-determination**, independent of rounds used, so it can group an accuracy-versus-rounds analysis without circularity.

2 Related Work

Existing turtle-soup benchmarks, TurtleBench [1], SPLAT [2] (whose interactive player–judge protocol we adopt), and TurtleSoup-Bench [3], score verification accuracy, solve efficiency, or faithfulness to the intended story. They score the destination; we score the trajectory. MUTATE [4] shows that frontier agents find far fewer mechanism-distinct paths than humans, and its diverge-then-narrow agent is a natural intervention to test against our metrics. The instrument itself adapts forward flow [7] and the Divergent Association Task [8] from human psychometrics, which treat semantic travel as a measure. The Small World of Words norms [9] provide the human anchor that ours is not yet (§3). Judge-bias catalogues [5, 6] shape the scoring: the model judge is confined to the one subscore that objective matching cannot reach.

3 Framework

3.1 Task and harness

A **Questioner** sees only the 汤面 and asks one yes/no question per round. An **Oracle** holds the 汤底 and answers 是/不是/与此无关. The Questioner may commit a final story at any round, or is forced to at the budget. The open-source harness has pluggable model providers and logs every round, with per-game seeds.

3.2 Puzzles

A puzzle nobody has solved carries no evidence that its clues suffice, that its solution is unique, or that the intended leap is reachable; an agent scoring zero on it teaches us nothing. Our set therefore takes only puzzles with a public record of human play: 22 items from a Chinese Turtle Soup community site, each keeping its source URL. Surfaces run 9–108 characters (median 34), solutions 13–200 (median 102), with 97 annotated key clues. Provenance is enforced in code, with a verified family, a quarantined generated family, and tests that fail if the two mix.

Each puzzle carries an **under-determination** score: how far the surface alone leaves a solver from the solution, measured by sampling twelve cold guesses from the surface with no Oracle feedback and taking the closest. It ranges 0.173–0.577 (median 0.347) over the set and is *not* explained by surface length ($r = -0.22$, n.s.). It therefore enters the mixed-effects analysis in §5 as a puzzle-level covariate.

3.3 Outcome scoring: composite, not surface

Judged outcome is **clue recall (70)** plus **causal-logic identity (30)**; Appendix B gives the detail. String matching alone cannot tell a wrong answer from a right one worded differently: we watched a frontier model reconstruct a hidden causal chain completely and score 0.00. Conversely, the clue half blocks "far = creative" gaming, because distance without validity scores zero.

3.4 Association-trajectory instrumentation

Each round, we extract the question's keywords, embed them, and aggregate them to a unit vector q_t. Two signals follow: the **stride** $s_t = d(q_t, q_{t-1})$ and the **anchor distance** $h_t = d(q_t, \mathcal{A})$. The hypothesized signatures are stalling ($\bar{s} \to 0$ with h flat) and drifting (normal $\bar{s}$ with h_t rising). The lexical `question_novelty` metric is kept as the ablation baseline that H2 predicts will miss synonym drift.

On the anchor, stated plainly. Ours is built from the puzzle's surface, solution, and clues; it is **not** a human association manifold, and we no longer describe it as one. Its terms overlap the clue set that defines the outcome. Rebuilding it from the surface alone also flips the drift slope's sign on a minority of traces (quantified in §5). We therefore report both anchors and treat a SWOW-based version [9] as the correct future form.

4 Experiment Designs

E1, round curve. Play to 30 rounds, forcing and scoring a checkpoint answer after each round.

E2, round budget. Hard caps $\{5, 10, 15, 20, 25, 30\}$ with a forced final answer.

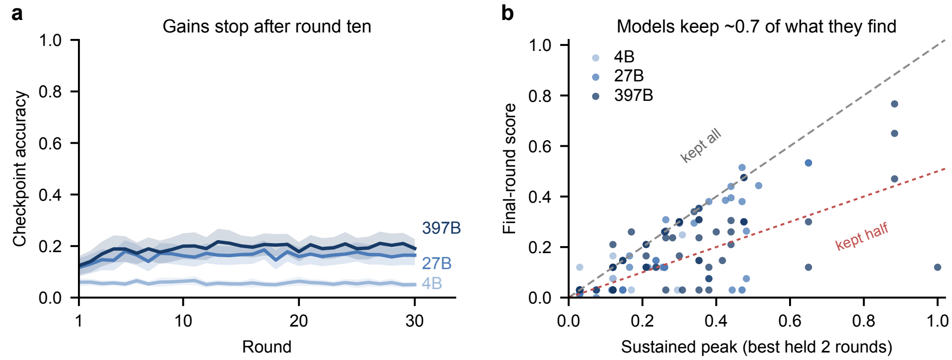

Figure 1: **(a)** Mean checkpoint accuracy by round (66 games per model; bands, 95% CI), full 0–1 range. **(b)** Final score against sustained peak (best held two rounds), one point per game; points under the red line ended below half their peak (18–20 of 66 per model).

E3, association trajectory. For every E1 game, compute $\{s_t\}$ and $\{h_t\}$ and test H1–H3, with the keyword extractor held fixed across models so no model is scored through its own vocabulary.

Prerequisite: the Oracle audit. The Oracle is the sole information channel: a weak one collapses answers to "irrelevant", after which no Questioner can succeed and every score measures the environment. Candidates are probed on held-out pairs before any agent claim, and **must come from a different model family than the Questioners**, or a Questioner's resemblance to the Oracle cannot be separated from its scale.

5 Results

All numbers come from one grid on the verified set: 22 puzzles $\times$ {Qwen3.5-4B, Qwen3.6-27B, Qwen3.5-397B-A17B; hereafter 4B, 27B, 397B} $\times$ 3 seeds, with a DeepSeek-V3.1 Oracle and logic judge (cross-family, audited at 95%; two logic samples per rating), and zero failed games. Accuracy denotes the composite score rescaled to $[0, 1]$. E1 comprises 198 thirty-round games with a scored checkpoint every round; E2, 1,188 games across caps $\{5, 10, 15, 20, 25, 30\}$.

5.1 More interaction stops helping after ten rounds (E1)

Checkpoint accuracy rises only in the first $\sim$10 rounds, and only for the larger models: 397B climbs from 0.124 (round 1) to 0.200 (round 10) and then stays between 0.18 and 0.21 through round 30, while 27B rises $0.117 \rightarrow 0.162$ and plateaus near 0.16. The 4B model is flat for the entire game, 0.05–0.06 at every checkpoint (Figure 1a). Twenty further rounds of evidence buy no accuracy at any scale.

Commitment separates the models more than accuracy does. 27B volunteers a final story in $11/66$ games (mean round 11.1), 397B in $9/66$ (14.2), and 4B in $2/66$ (23.5); the smallest neither improves nor concludes.

The plateau is an equilibrium of finding and losing, not stagnation. Each game's best checkpoint sits far above its last (Figure 1b). Because rewording alone can swing one checkpoint's score, we measure a *sustained* peak, the best value held over two consecutive checkpoints; even so, models end with only 0.69–0.76 of what they had demonstrably reached. Scale raises the sustained peak 3.5-fold ($0.084 \rightarrow 0.296$) but leaves retention unchanged: capability buys higher finds, not better keeping.

5.2 Budget pressure is asymmetric (E2)

End accuracy is insensitive to the round cap for the larger models (397B 0.19–0.21 and 27B 0.14–0.18 across all six caps), but **monotonically harmful for the smallest**: 4B falls from 0.050 at cap 5 to 0.004 at cap 30, with 175 of its 396 games ending by token-budget exhaustion rather than by answer (versus 2/396 for 27B and 0/396 for 397B). Longer games do not help a small model; they give it more rounds in which to fail.

The two score halves dissociate. Every model earns proportionally far more of the judged causal-logic half than the clue-recall half: 397B averages 12.6/30 logic against 7.2/70 clues, 27B 10.1 against 6.8, and 4B 2.9 against 0.2. Models reach the right causal shape well before, and often without, landing the annotated clue content. The two subscores therefore measure different competences; the logic half is not redundant with string-level recall.

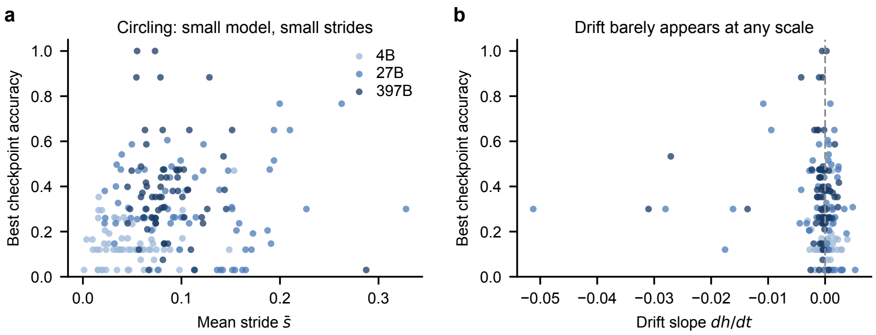

Figure 2: Per-trace geometry against best outcome ($n = 198$). **(a)** The models separate into stride regimes, but within a model stride does not predict outcome. **(b)** Drift slopes cluster near zero at every scale: H3's negative result and H2's absent signature in one view.

5.3 Two patterns: circling and abandoning (E3)

The models occupy different geometric regimes: mean stride is 0.049 for 4B against 0.088 (397B) and 0.114 (27B), and in the late game 4B's stride falls to 0.030, under half of 27B's 0.082, while it almost never commits. That is H1's signature, which we name *circling*: shrinking orbits of one hypothesis basin, with a sustained peak (0.084) showing the model never finds a solution to lose. The larger models fail in the opposite way, by *abandoning*: they find (sustained peak ≥ 0.5 in 4–6 of 66 games each) and then fail to keep or commit (§5.1). H2's drifting signature, by contrast, barely appears: drift slopes cluster near zero at every scale (Figure 2, right). The second real pattern is abandonment, which the hypothesized drift did not anticipate.

The same two patterns read differently in language space. Circling is only half-visible to lexical novelty: verbatim re-asks in late rounds are caught, but the same games earlier re-ask one hypothesis in fresh words, which lexical metrics score as new and only the geometry detects. Abandoning is invisible to lexical measures altogether; it shows in commitment behavior and checkpoint dynamics (§5.1), and Appendix B walks one game whose round-10 checkpoint story *is* the solution, later abandoned.

Pooled across models, stride correlates with best checkpoint score (Spearman $\rho = 0.23$, $n = 198$), but this is a between-model effect; **within** each model the correlation vanishes ($|\rho| \leq 0.16$ for every model and feature; Figure 2).

The mixed-effects test makes that precise. With puzzle as a random intercept and model tier as a fixed effect, adding the drift slope does not improve fit (LRT $p = 0.39$), nor does stride $\times$ tier ($p = 0.20$). Under-determination, the puzzle-level covariate, points the expected way (coefficient -0.031) but is not significant ($p = 0.22$) and removes only $\sim$8% of the puzzle variance, so the puzzle effect is not merely measured difficulty. **H3 is not supported within-puzzle**; trajectory geometry is an interpretive instrument, one that says *how* an agent fails, not a score predictor.

Both anchors, as promised (§3). Drift slopes from the solution-aware and surface-only anchors agree in rank ($\rho = 0.80$) yet flip sign on 38/198 traces, and the mixed-effects conclusion holds under both (surface-only LRT $p = 0.10$): the anchor changes traces, not the finding.

6 Limitations and Conclusion

A harness trap worth naming. Under too small a token budget, a reasoning model's logged "questions" are truncated chain-of-thought fragments to which every Oracle reply is "irrelevant"; from the transcript this is indistinguishable from an agent that cannot form a question, so decoding configuration is part of the environment.

Limitations. The anchor is not yet human-derived (§3), so H2's framing is descriptive until SWOW norms or collected human traces are in place; we do not equate human-unlike with wrong. Keyword extraction and encoders are themselves models; feedback incorporation, the most direct coordinate for separating circling from abandoning, is definable but unmeasured. The source site is public, so memorisation probes are required: the two puzzles rated easiest are also the two most widely circulated. Puzzles involve death and dark themes; content-flagged.

Conclusion. Turtle Soup turns hypothesis-space search into observable text, and trajectory geometry turns that text into behavioral signatures. An endpoint score projects the whole search onto one number, discarding exactly what improving an interactive agent requires: circling and abandoning agents need opposite interventions, and only the trajectory tells you which you have.

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A The puzzle set

Where the puzzles come from. A puzzle nobody has solved carries no evidence that its clues suffice, that its solution is unique, or that the intended leap is reachable; an agent scoring zero on it teaches us nothing. Our set is hand-picked from the classic repertoire of a Chinese Turtle Soup community: 22 puzzles people have played and solved, each keeping its source record. Every item was read and kept or rejected by hand; rejected were puzzles turning on deduction from stated evidence rather than a lateral reframing, and puzzles whose surfaces were incomplete at the source. Surfaces run 9–108 characters (median 34), solutions 13–200 (median 102).

How hard are they? Each puzzle carries an **under-determination** score: give a model the surface alone, with no Oracle and no feedback, let it write twelve complete stories, and take the closest to the real solution. The index is one minus that best guess, and is large when the surface leaves a solver far from the answer.

Under-determination	Puzzles
0.15–0.25	5
0.25–0.35	8
0.35–0.45	4
0.45–0.55	4
0.55–0.65	1

The median is 0.347, quartiles 0.273 and 0.410, full range 0.173–0.577: single-peaked with a thin hard tail. Easiest are the 海龟汤 story itself (0.17) and the desert matchstick (0.19). Hardest is a fourteen-character surface reading *our heights differ / a flowerpot broke / our heights are the same* (0.58).

Two properties make this usable as a covariate. It describes the puzzle, not any agent's performance on it. It is also independent of rounds used; a difficulty defined as "how many rounds this takes" would make an accuracy-versus-rounds plot confirm itself. Nor is it a proxy for surface length ($r = -0.22$, n.s.), so it carries information the obvious surrogate does not. It is one measure rather than two: a surface loose enough to admit many mechanisms is, for that same reason, one whose true mechanism takes more reframing to reach, so *how far* and *how many ways* are not separable here.

A caveat we report either way. The two puzzles scoring easiest are the two most widely circulated ones. Their surfaces may be easy to complete not because the inference is short but because the story is in the training data, in which case the measure reads familiarity as much as difficulty. A memorisation probe (asking for the solution with no surface given) separates the two and belongs in any use of this measure.

B Measuring agent performance

What counts as a correct answer. Judged outcome is **clue recall (70)** plus **causal-logic identity (30)**. Clue recall matches the answer against per-puzzle annotated key clues: objective and reproducible. The logic judge rates only whether cause → mechanism → outcome matches and is told explicitly to ignore wording. Because single ratings are unstable on borderline answers, ratings are sampled and averaged (two samples per rating in the reported grid). The gap between the two subscores is itself readable: right story, wrong vocabulary.

Clue matching tolerates paraphrase through character-bigram recall, which sets a floor on how short an annotated clue may be. Below that floor only a verbatim match counts, and the score reads vocabulary rather than content. Clues are annotated above the floor.

Does the logic half earn its thirty points? On the full grid, yes: the two halves dissociate for every model (§5), with each model earning a several-fold larger share of the logic half than of the clue half. If the two rankings had agreed, clue recall alone would have been the cleaner scale; they do not.

Where the accuracy curve comes from. E1 does not score once at the end. Every round, after the Oracle answers, the Questioner is required to commit its best complete story, and *that* is scored. Each point on the accuracy curve is therefore a full answer written with the evidence available at that round.

One game, worked. Puzzle `refsoup_021` (surface: a stormy night, a run-down temple, two photographs on the wall that seem to stare; by morning the narrator sees only two windows; solution: 墙上根本没有照片，是有人贴着窗户往里看), played by Qwen3.5-397B, seed 2. Table 1 shows six of thirty rounds; every cell is verbatim from the log, stories abridged to the clause carrying the hypothesis.

Three things the table shows that no mean curve can. **The checkpoint mechanism surfaces knowledge the agent never acts on.** The round-10 committed story *is* the solution, synthesized from earlier answers on a round whose own

轮	Question (abridged)	答	Committed story (abridged)	score (clue+logic)
1	照片里的是人吗?	是	那两张“照片”……是两个被钉死或封在墙洞里的人	0.12 (0+12)
3	主角第二天醒来时，还在那座寺庙里吗?	是	所谓的“两张照片”其实是两扇窗户……昏暗光线下……误看成了照片	0.91 (70+21)
10	“两扇窗户”是因为相框玻璃破碎脱落、变成通透的洞口吗?	不是	那根本不是照片，而是……躲在窗户后面（或贴在窗玻璃外侧）的人脸	**1.00** (70+30)
11	是因为照片被取走、露出了后面原本就存在的窗户吗?	不是	昨晚“盯着他的人像”，其实是窗外的人	**1.00** (70+30)
19	照片里的人像实际上是躲在照片后面的真人吗?	不是	窗户洞口被画着人像的布帘封住……雷雨导致遮挡物脱落	0.12 (0+12)
30	是因为积水使主角身体位置升高吗?	不是	两个被纸糊住的窗户洞口……雷雨导致纸张脱落	0.12 (0+12)

Table 1: Six rounds of one E1 game (`refsoup_021`, Qwen3.5-397B, seed 2): the Questioner's question, the Oracle's answer, the story the agent was forced to commit that round, and its composite score.

question was answered 不是. Yet the model never volunteers a final answer in all thirty rounds (one of the 57/66 non-committing 397B games), and by round 19 it has abandoned the correct reading for a falling-coverings mechanism it never escapes. **Stalling in fresh words happens at the largest scale too.** Rounds 22–30 are eight lexically distinct re-asks of the same flooding hypothesis, exactly the pattern lexical novelty misses. **The curve's round-to-round jitter is a scoring fact, not a knowledge fact.** The same understanding re-worded (rounds $3 \rightarrow 4$) swings the clue half $70 \rightarrow 0$. The mean curves of §5 average this out, which is why single-game curves are not the unit of analysis.